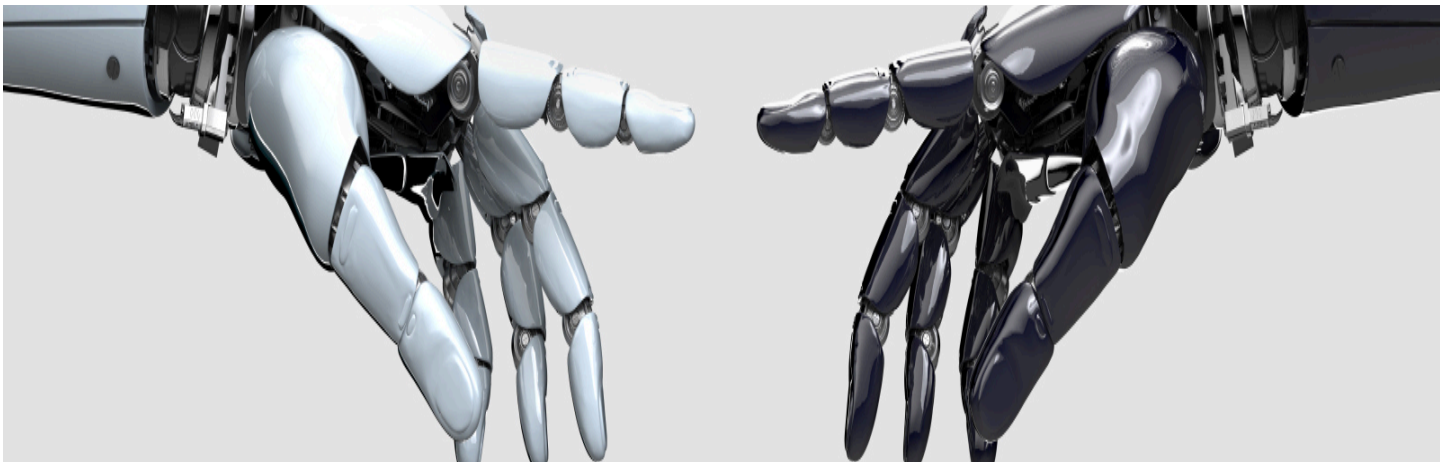

ZJUI Micro-minor Program in Intelligent Engineering

ZJUI_UndergradEd

6 分钟读完

In a world driven by rapid technological advancement, interdisciplinary collaboration, and dynamic problem-solving, Zhejiang University / University of Illinois Urbana-Champaign Institute (ZJUI) proudly introduce the ***Micro-minor Program in Intelligent Engineering (MPIE).*** This transformative education program, approved by Zhejiang University, is designed to prepare students for a future where intelligence systems are at the heart of global engineering challenges.

1.Overview

As a globally-oriented institute of higher education, ZJUI is committed to fostering interdisciplinary learning and innovative skill development. The **MPIE** is a forward-thinking initiative aimed at broadening the educational horizons of our undergraduate programs in Civil Engineering, Computer Engineering, Electrical Engineering and Mechanical Engineering.

MPIE has four focus areas that address complex engineering challenges:

- **Robotics and Automation**: This area covers advanced concepts in robotic systems, autonomous operation, and smart manufacturing, preparing students to excel in sectors such as industrial automation, healthcare, and service industries.

- **Artificial Intelligence (AI)**: From machine learning fundamentals to deep learning applications, students explore the capabilities of AI in transforming

industries, while addressing ethical and societal implications of AI technologies.

- **Data Science and Analytics**: With a focus on data-driven decision-making, this area provides hands-on training in big data analytics, statistical modeling, and data visualization, equipping students to tackle real-world data challenges in various engineering domains.

- **Smart Cities**: Students gain insight into sustainable urban development through technologies like IoT, smart infrastructure, and energy-efficient systems, aiming to build cities that are efficient, livable, and resilient.

MPIE enables students to supplement their primary engineering studies with a focus on emerging intelligent technologies, enhancing their competitiveness and adaptability in the global engineering landscape. By completing this program, students will not only develop a solid foundation in intelligent engineering principles but also gain practical skills that are highly sought after in the industry.

2. Requirements

To complete a **MPIE** program, students must fulfill the following:

1.1 Coursework (12 credits)

Students must complete a minimum of 12 additional credits in interdisciplinary engineering courses. These credits, detailed in Section 4 of the program curriculum, cannot be double-counted toward the requirements of both the student's primary and **MPIE** programs. However, up to 6 free elective credits may be applied.

1.2 Multi-disciplinary Senior Design Project or Thesis

Students are required to complete a senior design project or thesis that integrates principles across their primary program with MPIE focus areas. The design courses *ECE 445, ME 470 or ST 4001* may fulfill this requirement alongside primary graduation requirements.

Upon completion of these requirements, students will be awarded a certificate from Zhejiang University, signifying their proficiency in a **MPIE** area and their ability to apply core principles to solve complex problems. This certificate will also serve as evidence of their dedication and commitment to a rigorous and challenging program of study and will enhance their career prospects in the engineering industry.

3. Application

3.1 Eligibility

- **GPA Requirement**: Students must maintain a minimum cumulative GPA of 3.0 in both their major-specific courses and their overall coursework.

- **Coursework**: Eligible students should have completed specific courses from MPIE curriculum and are expected to fulfill all requirements by the end of the spring semester of their senior year.

3.2 Registration

Students interested in a **MPIE** program must submit an application form through the application portal during the Spring semester of their senior year. The application should include:

- A summary of relevant courses completed,
- A description of how their senior design project aligns with a **MPIE** focus area,

- A brief personal statement outlining their interest in the program.

3.3 Certification

The admission committee will assess applications based on academic performance, commitment to **MPIE**, relevant coursework and senior design or thesis project. Notifications regarding admission decisions will be sent out after evaluations are complete. Each student can only be admitted to one **MPIE** focus area. Qualified students will receive a micro-minor certificate of "Intelligent Engineering" in their chosen focus area from Zhejiang University, recognizing their interdisciplinary skills and expertise.

4. Curriculum

To fulfill the MPIE curriculum requirements, students may take advantage of course resources not only at ZJUI but also during exchanges at UIUC or at Zhejiang University's main campus. Additional courses not listed in the curriculum but deemed equivalent in content may be approved by ZJUI for credit toward the program requirements.

4.1 **Robotics and Automation**

The Robotics and Automation area is a highly interdisciplinary field combining core principles from mechanical engineering, electrical engineering, and computer science. This focus equips students with skills applicable across multiple sectors, including manufacturing, healthcare, and transportation. Courses in this area cover topics such as control systems, robotic kinematics, automation technology, and applications in real-world environments.

Course Code	Course Name	Credits	Prerequisite
ECE 470/ME 445	Introduction to Robotics	4	one of Math 225, Math 286, Math 415, Math 418
ECE 489/ME 446	Robot Dynamics and Control	4	ECE 470/ME 445
ECE 486	Control System	4	ECE 210
ECE 448/CS 440	Artificial Intelligence	3	CS 225; one of CS 361, STAT 361, ECE 313, MATH 362, MATH 461, MATH 463, STAT 400
ECE 449/CS 446	Machine Learning	3	CS 225, Math 257, ECE 313 or STAT 400
ECE 408	Applied Parallel Programming	4	ECE 220
ECE 484	Principles of Safe Autonomy	4	ECE 220, ECE 313
ECE 364	Programming Method for Machine Learning	3	Math 257
ME 440	Kinematics & Dynamics of Mechanical System	3	ME 370
ME 447	Computational Design and Dynamics of Soft Systems	4	CS 101, Math 257
CS 444	Deep Learning for Computer Vision	3	ECE 220
ECE 410	Neutral Circuits and System		
ECE 426/ABE 426	Principles of Mobile Robotics	4	Math 285, ECE 313 or IE300 or STAT 400, CS 101
ME 360	Signal processing	3.5	ME 340

**Courses highlighted in blue were already offered at ZJUI.*

4.2 **Artificial Intelligence**

The Artificial Intelligence (AI) area focuses on the creation and application of computer systems capable of performing tasks traditionally requiring human intelligence, such as image recognition, natural language processing, and complex decision-making. This interdisciplinary field integrates principles from computer science, mathematics, and statistics to equip students with skills in machine learning, neural networks, and AI ethics, preparing them to contribute to rapidly evolving AI applications across industries like healthcare, finance, and technology.

Course Code	Course Name	Credits	Prerequisite
ECE 448/CS 440	Artificial Intelligence	3	CS 225; one of CS 361, STAT 361, ECE 313, MATH 362, MATH 461, MATH 463, STAT 400
ECE 449/CS 446	Machine Learning	3	CS 225, Math 257, ECE 313 or STAT 400
ECE 364	Programming Method for Machine Learning	3	Math 257
ECE 410	Neural Circuits and Systems	3	MATH 285 or equivalent CS 101 or equivalent
ECE 417	Multimedia Signal Processing	4	ECE 310 or ECE 401; one of ECE 313, CS361, or STAT 400
ECE 418	Image & Video Processing	4	ECE 310; one of ECE 313, STAT 400, IE 300, Math 461; Math 257
ECE 421	Neural Interface Engineering	3	ECE 210 or BIOE 205 and NE 330

Course Code	Course Name	Credits	Prerequisite
CS 412	Introduction to Data Mining	3	CS 225
CS 443	Reinforcement Learning	3	CS 225, Math 241, Math 257, ECE 313
CS 447	Natural Language Processing	3	ECE 374
CS 444	Deep Learning for Computer Vision	3	Math 241, Math 257, ECE 313 or STAT 400

**Courses highlighted in blue were already offered at ZJUI.*

4.3 **Data Science and Analytics**

The Data Science and Analytics area focuses on employing statistical and computational methods to derive valuable insights from large, complex datasets. This interdisciplinary field merges principles from statistics, computer science, and engineering, equipping students with skills in data mining, predictive modeling, and data visualization. Students in this area are prepared to address data-driven challenges across sectors, including healthcare, finance, social media, and beyond, where informed decision-making is essential.

Course Code	Course Name	Credits	Prerequisite
ECE 365	Data Science and Engineering	3	ECE 313
ECE 364	Programming Method for Machine Learning	3	Math 257
CS 412	Introduction to Data Mining	3	CS 225
ECE 449/CS 446	Machine Learning	3	CS 225, Math 257, ECE 313 or STAT 400
ECE 491/CS 450	Numerical Analysis	3	CS 101, Math 257, Math 285
STAT 432	Basics of Statistical Learning	3	STAT 400
ECE 471	Data Science Analytics using Probabilistic Graph Models	3	ECE 313
ECE 408	Applied Parallel Programming	4	ECE 220
ME 453	Data Science in Manufacturing Quality Control	3	ME 270, Math 257, ECE 313 or STAT 400
CS 415	Data Visualization	3	CS 225

**Courses highlighted in blue were already offered at ZJUI.*

4.4 **Smart Cities**

The Smart Cities area centers on using technology and data to improve urban living by enhancing infrastructure, optimizing services, and minimizing environmental impact. This interdisciplinary field brings together principles from civil engineering, electrical engineering, and computer science, enabling students to work on sustainable urban development projects. Key topics include Internet of Things (IoT) integration, smart infrastructure, energy management, and environmental monitoring, all aimed at creating efficient, resilient, and livable cities for the future.

Course Code	Course Name	Credits	Prerequisite
CEE 340	**Energy and Global Environment**	**3**	**Phys 211: Phys 213 or ME 200; CEE 201 or IE 310; CEE 202 or STAT 200**
ECE 364	Programming Method for Machine Learning	3	Math 257
CEE 498/CS 498	Introduction to Smart Cities	3	
ECE 448/CS 440	Artificial Intelligence	3	CS 225; one of CS 361, STAT 361, ECE 313, MATH 362, MATH 461, MATH 463, STAT 400
CS437	Topics in Internet of Things	3	CS 241 or CS 341
ABE 436	Renewable Energy Systems	3	Phys 211
CEE 417	Urban Transportation Planning	3	
ECE 498DASC	Data Analytics for Smart Cities	3	

**Courses highlighted in blue were already offered at ZJUI.*

5. Appendix

The following is a list of prerequisites for the courses in **MPIE** curriculum. Some pre-requisites are optional. Please refer to the curriculum table for the corresponding program.

Course Code	Course Name	Credits
CS 101	Introduction to Computing: Engineering & Science	3
MATH 221	Calculus Ⅰ	4
MATH 231	Calculus Ⅱ	3
MATH 241	Calculus Ⅲ	4
MATH 257	Linear Algebra with Computational Application	3
MATH 285	Introduction to Differential Equation	3
PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec & Mag	4
CS 225	Data Structures	4
ECE 313/MATH 362	Probability with Engineering Application	3
CS 361/STAT 361	Probability & Statistics for Computer Science	3
MATH 461	Probability Theory	3
MATH 463/STAT 400	Statistics and Probability I	4
BIOE 310	Computational Tools for Biological Data	3
CEE 201	Systems Engineering & Economics	3
CEE 202	Engineering Risk & Uncertainty	3

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